basic education
Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

2023/24 ANNUAL TEACHING PLANS: ELECTRICAL TECHNOLOGY: ELECTRONICS: GRADE 12 (TERM 1)

TERM 1	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
CAPS TOPICS	**Safety (generic)**	**RLC**	**RLC**	**RLC**	**RLC**	**RLC**	**Semi-conductor devices**	**Semi-conductor devices**	**Semi-conductor devices**	**Semi-conductor devices** **Term 1 assessment**	**PAT consolidation and assessment**
TOPICS, CONCEPTS, SKILLS AND VALUES	- First aid - HIV, Aids awareness - OHS Act - Machine-specific safety measures	Effect of alternating current on R, L and C components in series and parallel circuits	- Inductive reactance $X = 2\pi fL$ - Capacitive reactance	- Impedance - Power - Phase angle - Power factor - Phasor and wave representation - Resonance - Q factor & bandwidth	**Calculations** - Series and parallel combination circuits containing ONE resistor, ONE capacitor and ONE inductor - Frequency changes - Phasor and wave representation - Resonance - Phasor diagram	- Phasor and wave representation - Resonance - Phasor diagram **Demonstration:** Show the effect of changing frequency in an RLC circuit forward resonance	**The field effect transistor** - Basic construction, symbols, operation, characteristics - Types of FET (NFET, JFET, MOSFET) - Characteristic curves & typical operating voltages, application as a switch - Application as an amplifier	**Uni-junction and Darlington transistor** - Basic construction, symbols, functional operation, characteristics - Characteristic curves & typical operating voltages - Application as a switch - Application as saw-tooth generator - Application as an amplifier	**Introducing integrated circuits** - Integrate circuits – the 741 op-amp - Basic construction, symbols, functional operation - Typical operating voltages - Characteristics of an ideal op-amp & application as an amplifier - Gain: Open loop and closed loop gain - Application as an inverting amplifier - Application as a non-inverting amplifier Calculations • Inverting amplifier $V_{out} = V_{in}(-\frac{R_f}{R_{in}})$ • Non-inverting amplifier ○ $V_{out} = V_{in}(\frac{R_f}{R_{in}} + 1)$ • Gain ○ $A_v = \frac{R_f}{R_{in}}$	**Practical:** Build a non-inverting amplifier on a breadboard using a 741 Op-Amp Use a function generator and oscilloscope to show input and output waveforms - Integrate circuits – the 555 timer ➢ Basic construction, symbols, functional operation ➢ Characteristics curves & typical operating voltages ➢ Application as a timer **Practical:** Build a clock pulse generator using a 555 timer IC on a breadboard and display the output on an oscilloscope	**PAT Simulation Complete** - **PAT design part 1: Circuit diagram** - **component list circuit description**
REQUISITE PRE-KNOWLEDGE	Introduction of the OHS Act, Electrical Machinery Regulations	The effect of AC on RLC series circuits				Introduction to semiconductor and solid-state devices					
RESOURCES (OTHER THAN TEXTBOOK) **TO ENHANCE LEARNING**	OHS Act - Safety signs in workshop First aid training manuals	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	RLC simulation "spook box"	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers FET and Darlington simulations	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	

TERM 1		WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
ASSESSMENT	**INFORMAL ASSESSMENT:** REMEDIATION	**Classwork, case studies, worksheets, homework, theory and practical, etc.**										
	SBA (FORMAL)	**PAT simulations 1 completed** Safe work practices are types of administrative controls that include procedures for safe and proper work used to reduce the duration, frequency, or intensity of exposure to a hazard. The section on tools and equipment must be infused when doing all PAT simulations.									**Assignment**	

2023/24 ANNUAL TEACHING PLANS: ELECTRICAL TECHNOLOGY: ELECTRONICS: GRADE 12 (TERM 2)

TERM 2	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
CAPS TOPICS	**Switching circuits**	**Switching circuits**		**Switching circuits**	**Switching circuits**	**Switching circuits**	**Switching circuits**	**Revision**	**June exam**	**June exam**	**June exam**
TOPICS, CONCEPTS, SKILLS AND VALUES	Principle of operation of switching **circuits using operational amplifiers and timers** - Multi-vibrators •Bistable multi-vibrator - Circuit diagram and operation - Measurement of input and output waveforms	**Mono-stable multi-vibrator** - Circuit diagram and operation - Measurement of input and output waveforms **Practical:** Construct a mono-stable amplifier on a breadboard using a 741 Op- Amp, 555 timer and LEDs **Astable multi-vibrator** - Circuit diagram and operation - Measurement of input and output waveforms		**Practical:** Construct an astable amplifier on a breadboard using a 741 Op-Amp, 555 timer and show output using LEDs and the oscilloscope **Schmidt Trigger** - Circuit diagram and operation - Display the input waveform in relation to the output waveform on the oscilloscope	**Practical:** Construct a Schmidt Trigger on a breadboard using a 741 Op-Amp **Comparator and summing amplifier** - Circuit diagram and operation - Display the input waveform in relation to the output waveform on the oscilloscope - Calculations, etc	**Practical:** Construct a comparator on a breadboard using a 741 Op-Amp **Practical:** Construct a summing amplifier on a breadboard using a 741 Op-Amp	**Differentiator and integrator** - Circuit diagram and operation - Display the input waveform in relation to the output waveform on the oscilloscope - Influence of time constant on the output waveform	Revision term 1 & 2 content			
REQUISITE PRE-KNOWLEDGE	Electronic components and how they work										
RESOURCES (OTHER THAN TEXTBOOK) **TO ENHANCE LEARNING**	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers 741 Op-Amp and 555 IC simulations		Educational videos and IT-related resources Old question papers 741 Op-Amp and 555 IC simulations	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers 741 Op-Amp and 555 IC simulations	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers		
ASSESSMENT **INFORMAL ASSESSMENT:** REMEDIATION	**Classwork, case studies, worksheets, homework, theory and practical, etc.**										
SBA (FORMAL)	**PAT simulations 2 completed** Safe work practices are types of administrative controls that include procedures for safe and proper work used to reduce the duration, frequency, or intensity of exposure to a hazard. The section on tools and equipment must be infused when doing all PAT simulations.									**Mid-year examination**	

2023/24 ANNUAL TEACHING PLANS: ELECTRICAL TECHNOLOGY: ELECTRONICS: GRADE 12 (TERM 3)

TERM 3	WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10	WEEK 11
CAPS TOPICS	**Amplifiers**	**Amplifiers**	**Amplifiers**	**Amplifiers**	**Amplifiers**	**Revision**	**Prep exam PAT mod**	**Prep exam PAT mod**	**Prep exam PAT mod**	**Prep exam PAT mod**	**Prep exam PAT mod**
TOPICS, CONCEPTS, SKILLS AND VALUES	**Amplifier theory** - Determination of a typical load line by means of Ohm's Law (revision) - Basic concept of class A, B and C amplifiers AB - Principles of negative feedback, the decibel and log	**Resistor capacitor coupled amplifier (NPN transistor)** - Basic operation - Circuit diagram & practical - Input and output curves - Frequency response curve - Gain & loss in decibel calculations	**Transformer coupled amplifier (NPN transistor)** - Basic operation - Circuit diagram - Input and output curves - Frequency response curve	**Push pull amplifier (NPN, PNP transistor)** - Basic operation - Circuit diagram & practical - Input and output curves - Frequency response curve - Gain & loss in decibel calculations - Typical biasing	**Radio frequency amplifier** - Basic operation - Circuit diagram - Input and output curves - Frequency response curve - Typical biasing **Practical:** Construct a simple RF amplifier	Revision Term 1, 2 & 3 content					
REQUISITE PRE-KNOWLEDGE	Operating principle of the transistor as an amplifier										
RESOURCES (OTHER THAN TEXTBOOK) **TO ENHANCE LEARNING**	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers						
ASSESSMENT **INFORMAL ASSESSMENT:** REMEDIATION											
ASSESSMENT **SBA (FORMAL)**	**PAT project** Safe work practices are types of administrative controls that include procedures for safe and proper work used to reduce the duration, frequency, or intensity of exposure to a hazard. The section on tools and equipment must be infused when doing all PAT simulations.						**Preparatory exam**				

2023/24 ANNUAL TEACHING PLANS: ELECTRICAL TECHNOLOGY: ELECTRONICS: GRADE 12 (TERM 4)

TERM 4		WEEK 1	WEEK 2	WEEK 3	WEEK 4	WEEK 5	WEEK 6	WEEK 7	WEEK 8	WEEK 9	WEEK 10
CAPS TOPICS		**Amplifiers** **SBA moderation**	**Amplifiers** **SBA moderation**	**Amplifiers** **SBA moderation**	**Revision** **SBA moderation**	**NCS Exams**	**NCS Exams**	**NCS Exams**	**NCS Exams**	**NCS Exams**	**NCS Exams**
TOPICS, CONCEPTS, SKILLS AND VALUES		**Hartley Oscillator (NPN or FET transistor)** - Basic operation - Circuit diagram - Output waveform - Tank circuit	**Colpitts Oscillator (NPN or FET transistor)** - Basic operation - Circuit diagram - Output waveform - Tank circuit	**RC Phase Shift Oscillator (NPN or FET Transistor**) - Basic operation - Circuit diagram - Output waveform - Tank circuit **Practical:** Construct an RC phase shift oscillator on a breadboard and show the output wave on an oscilloscope						
REQUISITE PRE-KNOWLEDGE		Operating principle of the transistor as an amplifier									
RESOURCES (OTHER THAN TEXTBOOK) **TO ENHANCE LEARNING**		Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers	Educational videos and IT-related resources Old question papers							
ASSESSMENT	**INFORMAL ASSESSMENT:** REMEDIATION	**Classwork, case studies, worksheets, homework, theory and practical, etc.**									
	SBA (FORMAL)	Final examination									